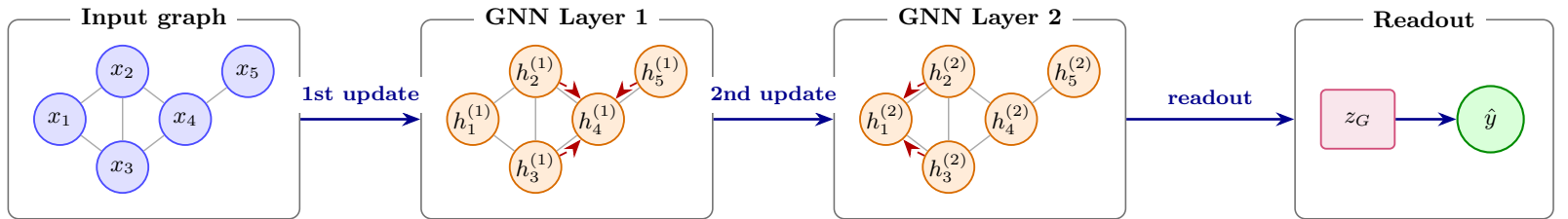


# Graph Neural Network: Message Passing Schematic



Node features  
 $x_v \in \mathbb{R}^{d_0}$

$$h_v^{(1)} = \sigma \left( W^{(1)} x_v + \sum_{u \in \mathcal{N}(v)} M^{(1)} x_u \right) \quad h_v^{(2)} = \sigma \left( W^{(2)} h_v^{(1)} + \sum_{u \in \mathcal{N}(v)} M^{(2)} h_u^{(1)} \right)$$

$$z_G = \text{READOUT}(\{h_v^{(2)}\}_{v \in V})$$

$$\hat{y} = f_\theta(z_G)$$

**Legend:**



input feature



hidden state



prediction



graph edge



message